FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2019

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

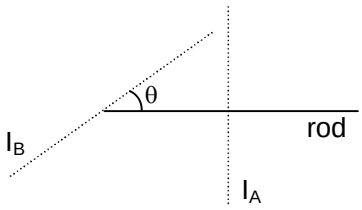
SECTION – A

1. ACD

Sol. For the angular momentum to remain conserved, the final velocity of the particle must not be zero.

2. ABC

Sol. $I_B = \frac{m\ell^2}{3}\sin^2\theta,\ I_A = \frac{m\ell^2}{12}$



3. A

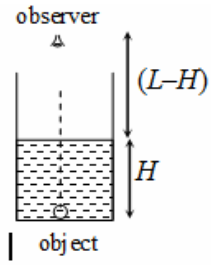
Sol. $X_{app} = L - H + \frac{H}{\mu}$

$$\frac{dX_{app}}{dt} = \frac{dH}{dt}\left(\frac{1-\mu}{\mu}\right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = -\frac{2H}{r}\frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{app}}{dt} = \frac{2KV}{\pi r^3}\left(\frac{(1-\mu)}{\mu}\right)$$



4. AC

Sol. $\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B$

at, $t = t_0$, $\frac{dN_B}{dt} = 0$, $N_B = \frac{3N_0}{2}$

$\Rightarrow 2N_0\lambda_1 e^{-\lambda_1 t_0} - \lambda_0 \frac{3N_0}{2} = 0$

$t_0 = \frac{1}{\lambda_1} \ln \frac{4}{3}\frac{\lambda_1}{\lambda_2}$

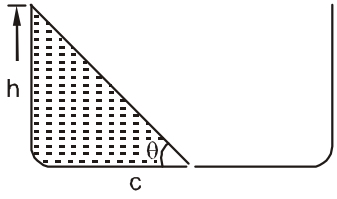
5. AC

Sol. The maximum amount of water that can be retained is shown in the fig. If θ is the angle made by the water surface with the horizontal then.

$\tan\theta = \frac{h}{c} = \frac{a}{g} \quad \Rightarrow a = \frac{hg}{c}$

So the maximum volume that can be retained is $\frac{1}{2} \times h \times c \times b$.

And $F = \left[M + \frac{hcb\rho}{2}\right]\frac{hg}{c}$



6. ABCD

Sol. Since both the gases are contained in the same vessel, temperature of both the gases is same.

Average KE per molecule of a diatomic gas is 5/2 KT. Hence, average KE permolecule of both the gases is same. Therefore, option (a) is correct.

$\nu_{rms} = \sqrt{\frac{3RT}{M}}$

Hence, $\frac{(\nu_{rms})_2}{(\nu_{rms})_1} = \sqrt{\frac{M_1}{M_2}} = \sqrt{16} = 4$

Hence, option(b) is correct.

Let molar mass of B be M, then that of A will be equal to 16 M.

Let mass of gas B in the vessel be m, then that of A will be 2m. The number of moles of a gas, in the vessel will be n = m/M. hence, number of moles of gases A and B will be

$n_1 = \frac{2m}{16M}$ and $n_2 = \frac{m}{M}$

Hence, $\frac{n_1}{n_2} = \frac{1}{8}$

Hence, option(d) is correct.

Partial pressure exerted by a gas is

$P = \frac{nRT}{V}$

Hence, $\frac{P_2}{P_1} = \frac{n_2}{n_1} = 8$

Therefore, option(c) is also correct.

7. B

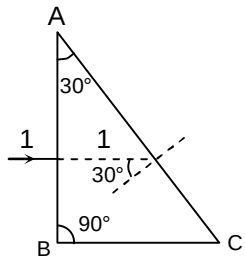
Sol. (P) Refractive index of the prism is the minimum value required for ray (1) to undergo total internal reflection at face AC. Ray (1) falls on face AC at an angle of incidence 30°

$\therefore$ $30° > i_C$

$\sin 30° > \sin i_C$

$\therefore$ $\mu > 2$

Minimum value of μ can be taken as 2.



(Q) For ray 2, refractive angle of prism is 30°. Apply Snell's law for refraction at face AB.

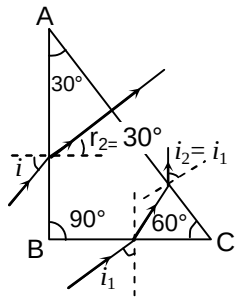
$1 \sin i = \mu \sin r$

$i = 90°$

(R) Using the relation $i_1 + i_2 = A + \delta$ for ray 2.

$90° + 0° = 30° + \delta$

$\delta = 60°$



(S) $\mu = \dfrac{\sin\left(\dfrac{A+\delta m}{2}\right)}{\sin\dfrac{A}{2}} \Rightarrow \delta m = 120°$

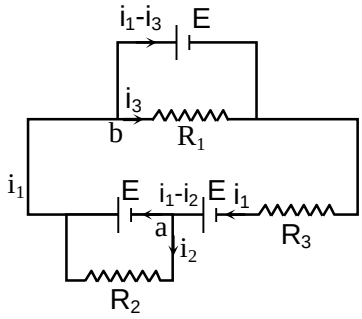
8. D

Sol. On solving by KVL we get, $i_1 = \dfrac{E}{R}$, $i_2 = -\dfrac{E}{R}$, $i_3 = \dfrac{E}{R}$

$\Rightarrow$ $i_1 - i_2 = \dfrac{2E}{R}$ and $i_1 - i_3 = 0$

So reading of ammeter A_1 is $E/R = 4/3$

A_2 is $2E/R = 8/3$, A_3 is zero and $\Delta V_{ab} = 4V$



9. A

Sol. RC is time constant $\sqrt{LC}$ is inverse of frequency and $\dfrac{Pc}{Q}$ is dimensionless.

10. B

Sol. For closed pipe, $\nu = \dfrac{(2n+1)v}{4L}$ and for first overtone, n = 1

For open pipe, $\nu = \dfrac{nv}{2L}$ and for fourth harmonic, n = 4.

SECTION – D

11. 00000.50

Sol. The apparent weight of block and apparent weight of fluid are changed by same amount.

12. 00002.40

Sol. $V_x = 18 \cos 60° = 9$ m/s.

$V_{1y} = \sqrt{15^2 - 9^2} = 12\,m/s$ $V_{2y} = -12$ m/s

$\therefore \quad \Delta t = \dfrac{-12-(12)}{-10} = \dfrac{12}{5}\sec.$

13. 00200.00

Sol. No relative motion between source and observers.

14. 00011.00

Sol. $I_1 = \left(\dfrac{s}{s+99}\right) \times I$

$\dfrac{I_1}{I} = \dfrac{s}{99+s} = 0.1$

$s = 11\Omega$

15. 00003.75

Sol. $T = \dfrac{3}{8} M\,\omega^2 L^2$

Stress $= \dfrac{F_\perp}{(A/\cos\theta)} = \dfrac{3}{8}\dfrac{M\omega^2 L^2 \cos^2\theta}{A}$

16. 00004.00

Sol. Newton's law for cooling.

Rate of decrease of temperature $\propto$ temperature difference between object and surrounding.

$\dfrac{75-65}{2}$ = Ab (70 – 30)

$\dfrac{55-45}{t}$ = Ab (50 – 30)

$\therefore$ t = 4 min.

17. 00002.17

Sol. $\dfrac{2.17+2.17+2.18}{3} = 2.1733 \ldots \approx 2.17$ mm upto 3 significant figure.

18. 00000.89

Sol. $2g - T = 2a$ …(i)

$TR = I\alpha$ …(ii)

$a = R\alpha$ …(iii)

From (ii) and (iii) $T = \dfrac{Ia}{R^2}$

$\therefore \quad 2g = a\left(2 + \dfrac{I}{R^2}\right)$

$\Rightarrow \quad a = \dfrac{2g}{\left(2+\dfrac{I}{R^2}\right)} = \dfrac{2\times 9.8}{2+\dfrac{0.2}{0.01}} = \dfrac{9.8}{11} \approx 0.89\ m/s^2$

Chemistry

PART – II

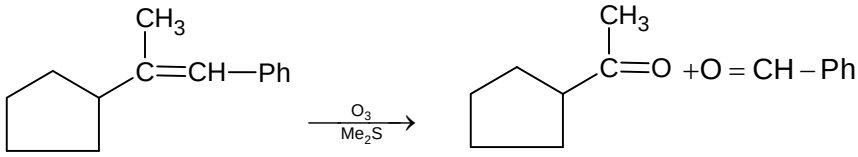
SECTION – A

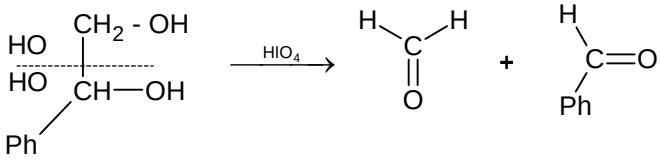
19. BCD

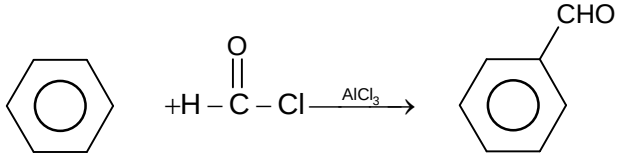
Sol. C is acid and B and D contain α - H w.r.t. $\diagdown C{=}O \diagup$ group.

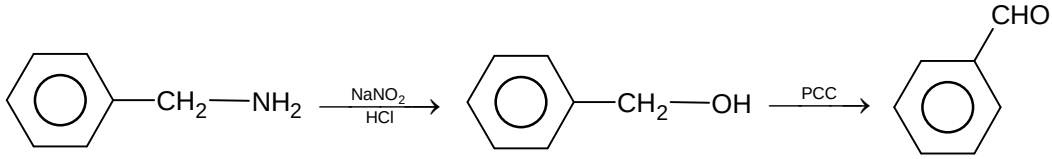
20. ABCD

Sol.

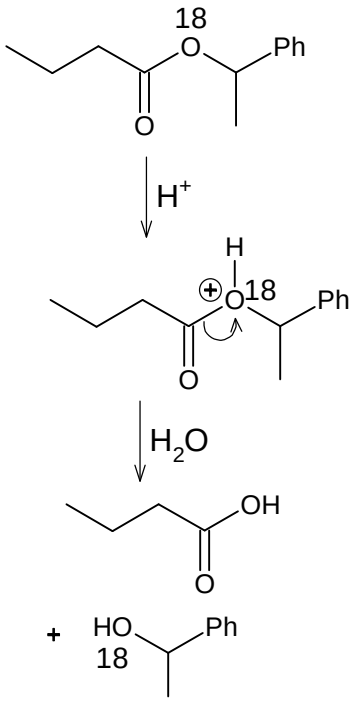
(A) 

(B) 

(C) 

(D) 

21. AC

Sol. 

22. ABC
Sol. Self explanatory reactions.

23. BCD
Sol. All the reagents in B, C and D are suitable.

24. ABC

25. B
Sol. M – H length = $H_2Te > H_2Se > H_2S > H_2O$
HMH B.A. = $H_2O > H_2S > H_2Se > H_2Te$
B.P. = $H_2O > H_2Te > H_2Se > H_2S$
(due to hydrogen bonding in H_2O)
All have two L.P. on central atom.

26. D
Sol.

	Ag^+	Zn^{2+}	Mg^{2+}	Ba^{2+}
Cl^-	White ppt.	No ppt.	No ppt.	No ppt.
S^{2-}	Black ppt.	White ppt.	No ppt.	No ppt.
CO_3^{2-}	Pale yellow ppt.	White ppt.	White ppt.	White ppt.
SO_4^{2-}	White ppt. which is sparingly soluble	No ppt.	No ppt.	White ppt.

In code (D) all are given with correct observation.

27. D
Sol. $\overset{IV}{PbO_2} + H_2SO_4 \xrightarrow{\Delta} \overset{II}{PbSO_4} + O_2 + 2H_2O$

$Na_2S_2O_3 + 5H_2O + 4Cl_2 \longrightarrow 2NaHSO_4 + 8HCl$

$N_2H_4 + 2I_2 \longrightarrow N_2(g) + 4HI$

$XeF_2 + 2NO \longrightarrow Xe(g) + 2NOF$

28. B
Sol. $Cu^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$
$Cd^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10}$
$Fe^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
$Mn = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

SECTION – D

29. 00004.00
Sol. CMC of soap
10^{-4} M to 10^{-3} M
x = 4

30. 00002.92
Sol. $\lambda = \frac{1}{t_{avg}} = \frac{1}{t}\ln\frac{[A_0]}{[A_t]}; \frac{1}{7.2} = \frac{1}{t}\ln\frac{100}{100 - \frac{100}{3}}$

$t = 2.303 \times 7.2 \times 0.1761$

31. 00000.02

Sol. $C = \dfrac{0.16/32}{500/1000}M$

$\alpha = \sqrt{K_b / c} = 2 \times 10^{-2}$

32. 00000.33

Sol. For ideal gas mole % = volume %

$$\underset{\text{initial moles}}{\underset{\text{at eqm}}{}} \quad \underset{\underset{a(1-\alpha)}{a}}{N_2O_2(g)} \rightleftharpoons \underset{\underset{2a\alpha}{0}}{2NO_2(g)}$$

As per given, original volume

$= \dfrac{75}{100} \times$ Volume at equilibrium at constant T and P $(V \propto n)$

$\therefore$ $a = 0.75 \times a(1 + \alpha)$

$\Rightarrow$ $\alpha = 0.33$

33. 00006.50

Sol. $pH = \dfrac{1}{2}\left(pK_w + pK_a - pK_b\right) = 6.5$

34. 00080.32

Sol. $\Delta H = \Delta E + \Delta n_g RT$

$\Rightarrow$ $82.8 = \Delta U + \dfrac{1 \times 8.314 \times 298}{1000}$

$\Rightarrow$ $\Delta U = 80.32$

35. 01129.05

Sol. $\Delta_r G° = \Delta_r H° - T \times \Delta_r S°$

$\Delta_r S° = 2 \times 81 - 4 \times 24 - 3 \times 205$ J/mol

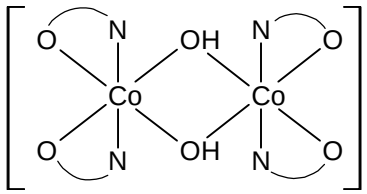
$\therefore$ $\Delta_r H° = -2258.1$ kJ/mol

$\Delta_r H° = 2 \times \Delta_f H°\ (Cr_2O_3, s)$

$\therefore$ $\Delta_f H°\left(Cr_2O_3, s\right) = -\dfrac{2258.1}{2} = -1129.05$ kJ / mol

36. 00004.00

Sol.



$\left(Gly^{-1} = NH_2 - CH_2 - COO^-\right)$

$= \left(\ddot{N} - \bar{O}\right)$

Mathematics

PART – III

SECTION – A

37. ABCD

Sol. Let $z_1 = \cos\theta + i\sin\theta$ and $z_2 = \cos\phi + i\sin\phi$

$a = \cos\theta, b = \sin\theta$

$c = \cos\phi,\ d = \sin\phi$

$z_1\bar{z}_2 = \cos(\theta - \phi) + i\sin(\theta - \phi)$

$\text{Re}(z_1\bar{z}_2) = 0 \Rightarrow \cos(\theta - \phi) = 0$

$\Rightarrow \quad \theta - \phi = \frac{\pi}{2}$

$\therefore \quad z_1 = -\sin\phi + i\cos\phi$

$\Rightarrow \quad a = -\sin\phi = -d,$

$b = \cos\phi = c$

Now, $w_1 = a + ic = -\sin\phi + i\cos\phi$

$w_2 = b + id = \cos\phi + i\sin\phi$

$|w_1| = |w_2| = 1 \Rightarrow |w_1 w_2| = 1$

Also, $\text{Re}(\bar{w}_1 w_2) = 0$

38. ABCD

Sol. $\alpha + \beta + \gamma + \delta = p$ and $\alpha\beta\gamma\delta = s$

By using A.M $\geq$ G.M. inequality

$\therefore \frac{p}{4} \geq \sqrt[4]{s}$

$\therefore \ p^4 \geq 256s$

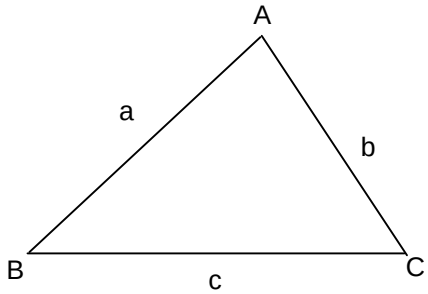
Similarly we can evaluate other options by same inequality

39. BC

Sol. $\cos B + \cos C = 4\sin^2\left(\frac{A}{2}\right)$

$\Rightarrow \quad 2\cos\frac{B+C}{2}\cos\frac{B-C}{2} = 4\sin^2\frac{A}{2}$

$\Rightarrow \quad 2\cos\left(\frac{\pi}{2} - \frac{A}{2}\right)\cos\frac{B-C}{2} = 4\sin\frac{A}{2}\sin\left(\frac{\pi}{2} - \frac{B+C}{2}\right)$



$$\Rightarrow \quad \frac{\cos\frac{B-C}{2}}{\cos\frac{B+C}{2}}=2 \quad \Rightarrow \quad \frac{\cos\frac{B}{2}\cos\frac{C}{2}+\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}\cos\frac{C}{2}-\sin\frac{B}{2}\sin\frac{C}{2}}=2$$

$$\Rightarrow \quad \tan\frac{B}{2}\tan\frac{C}{2}=\frac{1}{3} \Rightarrow \quad \frac{\Delta}{s(s-b)}\frac{\Delta}{s(s-c)}=\frac{1}{3}$$

$$\Rightarrow \quad 3s(s-a)(s-b)(s-c)=s^2(s-b)(s-c)$$

$$\Rightarrow \quad 3(s-a)=s \Rightarrow 2s=3a$$

$$\Rightarrow \quad a+b+c=3a \qquad \Rightarrow b+c=2a$$

$$\Rightarrow \quad AB+AC=2a$$

Also $AB+AC>BC=a$

So locus of A is an ellipse with foci at B and C and the length of the major major axis in 2a.

40. AC

Sol. The function f is no differentiable at x = 0, $x=\frac{\pi}{2}$ as

$$f'(0-)=-10,\ f'(0+)=1;\ f'\left(\frac{\pi}{2}-\right)=1,\ f'\left(\frac{\pi}{2}+\right)=-1$$

The function $f'(x)$ is given by $f'(x)=\begin{cases}3x^2+2x-10 & -1\le x<0\\ \cos x & 0<x<\frac{\pi}{2}\\ -\sin x & \frac{\pi}{2}<x\le\pi\end{cases}$

The critical points of f are given by $f'(x)=0$ or $x=0,\frac{\pi}{2}$, Thus critical points are $x=\frac{\pi}{2}$, x = 0. Since $f'(x)>0$, for $0<x<\frac{\pi}{2}$ and $f'(x)<0$, for $\frac{\pi}{2}<x<\pi$ so f has local maxima at $x=\frac{\pi}{2}$. Also $f'(x)<0$ for $-1\le x<0$ and $f'(x)>0$ for $0<x<\frac{\pi}{2}$ so f has local minima at x = 0. Since $f(-1)=-9$, $f\left(\frac{\pi}{2}\right)=1$, $f(0)=0$ and $f(\pi)=0$. Thus f has absolute minimum at $x=-1$ and absolute maximum at x = 0.

41. ABC

Sol. $I_n=\int_{-\pi}^{\pi}\frac{\sin nx}{(1+\pi^x)\sin x}dx$

$$=\int_0^{\pi}\left(\frac{\sin nx}{(1+\pi^x)\sin x}+\frac{\pi^x\sin nx}{(1+\pi^x)\sin x}\right)dx=\int_0^{\pi}\frac{\sin nx}{\sin x}dx$$

Now, $I_{n+2}-I_n=\int_0^{\pi}\frac{\sin(n+2)x-\sin nx}{\sin x}dx$

$$= \int_0^{\pi} \frac{2\cos(n+1)x.\sin x}{\sin x}dx = 0$$

$$\Rightarrow I_1 = \pi,\ I_2 = \int_0^{\pi} 2\cos x\,dx = 0$$

42. AD

Sol. $f(x) = \int_0^x (1+|1-t|)dt,\ x > 2$

$$= \int_0^1 (2-t)dt + \int_1^x t\,dt = \frac{1}{2}x^2 + 1$$

We have, $f(x) = \begin{cases} \frac{1}{2}x^2 + 1, & x > 2 \\ 5x + 1, & x \le 2 \end{cases}$

$\therefore \lim\limits_{x\to 2^+} f(x) = 3$ and $\lim\limits_{x\to 2^-} f(x) = 11$

Hence, f(x) is not continuous at x = 2.

$$Rf'(2) = \lim_{h\to 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{1}{2}(2+h)^2 + 1 - 11}{h}$$

$$= \lim_{h\to 0} \frac{(2+h)^2 - 20}{2h} \to \pm\infty$$

43. A

Sol. (P) $a_1 = \int_0^{\pi/2} dx = \frac{\pi}{2},\ a_2 = \pi$

Also, $a_{n+2} + a_n - 2a_{n+1}$

$$= \int_0^{\pi/2} \frac{2\cos(2n+2)x - \cos(2n+4)x - \cos(2nx)}{1-\cos 2x}dx$$

$$= \int_0^{\pi/2} \frac{2\cos(2n+2)x - 2\cos(2n+2)x\cos 2x}{1-\cos 2x}dx \quad = \int_0^{\pi/2} 2\cos(2n+2)x\,dx$$

$$= \frac{2}{2n+2}\left(\sin(2n+2)x\right)_0^{\pi/2} = 0$$

$\therefore$ $a_1, a_2, a_3, \ldots$ are in A.P.

(Q) $I_{n+1} + I_{n+3}$

$$= \int_0^{\pi/4} \tan^{n+1}x \sec^2 x\,dx$$

$$= \frac{1}{n+2}\tan^{n+2}x\Big]_0^{\pi/4} = \frac{1}{n+2}$$

$\therefore a_n = n+2$

$\Rightarrow a_1, a_2, \ldots$ are in A.P.

(R) $I_n - I_{n-1} = \int_0^{\pi/2} \frac{\sin^2(nx) - \sin^2(n-1)x}{\sin x} dx$

$= \int_0^{\pi/2} \frac{\sin(2n-1)x \sin x}{\sin x} dx$

$= -\frac{1}{2n-1}\cos(2n-1)x\Big]_0^{\pi/2} = \frac{1}{2n-1}$

$\Rightarrow a_1, a_2, a_3, \ldots$ are in H.P.

(S) $b_n = \int_0^{\pi/2} \frac{1-\cos 2nx}{1-\cos 2x} dx = \frac{n\pi}{2}$ (see (a))

$a_n = \frac{\pi}{2} \forall n$

$\therefore$ $a_1, a_2, a_3, \ldots$ are in A.P., G.P. and H.P.

Also $a_1, a_2, a_3, \ldots$ is a constant sequence.

44. C

Sol. Let $A = \{a_1, a_2, \ldots a_n\}$. For each $a_i \in A$, we have following four choices

(1) $a_i \in P$, $a_i \in Q$

(2) $a_i \in P$, $a_i \notin Q$

(3) $a_i \notin P$, $a_i \in Q$

(4) $a_i \notin P$, $a_i \notin Q$

For $P \cup Q = A$ or $P \cap Q = \varphi$ the number of choice is 3^n. The singleton can be chose in ${}^nC_1$ ways, rest of the elements in 3^{n-1} ways. $P \cup Q = \varphi$ in just one way.

45. A

Sol. (P) $z = e^{i\theta}$

$= \text{Re}\left(\frac{2ie^{i\theta}}{1-e^{i2\theta}}\right)$

$= \text{Re}\left(\frac{2ie^{i\theta}}{2\sin^2\theta - i2\sin\theta\cos\theta}\right) = \text{Re}\left(-\frac{1}{\sin\theta}\right) = (-\infty, -1] \cup [1, \infty)$

(Q) $3^x = t$

$-1 \le \frac{8t}{9-t^2} \le 1$

$0 \le \frac{t^2 - 8t - 9}{t^2 - 9}$

$t < 3 \cup 9 \le t$

$x < 1 \cup 2 \le x$ ……..(i)

where $\frac{8t}{9-t^2} \le 1$

$\frac{(t+9)(t-1)}{(3-t)(3+t)} \le 0$

$t \le 1 \cup 3 < t$

$x \le 0 \cup 1 < x$ .....(ii)

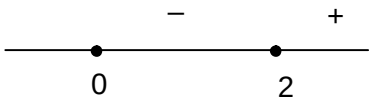
from (i) and (ii)

$x \le 0 \cup 2 \le x$

(R) $f(\theta) = 2\sec^2\theta \ge 2$

(S) $f'(x) = x^{3/2} \times 3 + (3x-10)\frac{3}{2}x^{1/2}$

$= 3\sqrt{x}\left(x + \frac{3x-10}{2}\right)$



$x \ge 2$

46. A

Sol. a, b, c are three roots of the quadratic equation

$\left(\left(4f(-1)-3\right)x^2 + \left(4f(1)-3\right)\right)x + f(2) = 0$

$\Rightarrow 4f(-1) = 3, 4f(1) = 3$ and $f(2) = 0$

Let $f(x) = (x-2)(Ax+B)$

Now, $3 = 4f(-1) = 4(-3)(-A+B) \Rightarrow A - B = \frac{1}{4}$

$3 = 4f(-1) = 4(-1)(A+B) \Rightarrow A + B = \frac{-3}{4}$

$\therefore\ A = \frac{-1}{4}, B = \frac{-2}{4}$

Thus, $f(x) = \frac{1}{4}\left(4 - x^2\right)$.

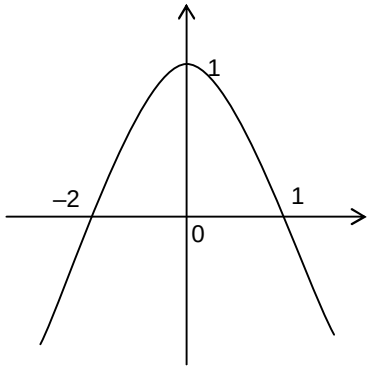
The graph of $y = f(x)$ is given in Figure.

(A) x coordinates of points of intersection of $y = f(x)$ when the x – axis are ± 2.

(B) Area $= \frac{3}{2}\int_{-2}^{2}\frac{1}{4}\left(4-x^2\right)dx = \frac{3}{8}(2)\left(4x - \frac{x^2}{3}\right)\Big]_0^2 = 4$

(C) Maximum value of f(x) is 1

(D) Length of intercept on the x – axis is 4.



SECTION – D

47. 67676.00

Sol. $x_{51} = x_{50} + 51^2$

$= \frac{1}{2}(50)(51)^2 + 51^2 = 67626$

48. 00240.00

Sol. We must have $x \geq 1$ and $y \geq 2$, since x is less than y. That is, y = j with $2 \leq j \leq 9$, then x can take the j – 1 values from 1 to j – 1, and z can take the j values from 0 to j – 1. For each value y = j, therefore, xyz can take j (j – 1) values. We can now get the answer by summing over all values of y:

$\sum_{j=2}^{9} j(j-1) = \sum_{j=1}^{9} j(j-1) = \sum_{j=1}^{9} j^2 - \sum_{j=1}^{9} j$

$= 285 - 45 = 240$

49. 01300.00

Sol. $b = \sum_{r=1}^{50} a_r$

But $a_r = r^2 \frac{(50-r+1)}{r} = \sum r(51-r)$

50. 00004.00

Sol. $\cos\beta = \frac{a^2 + 16 - 8}{2 \times a \times 4}$

$\Rightarrow a^2 - 4\sqrt{3}a + 8 = 0$

$\Rightarrow a_1 + a_2 = 4\sqrt{3},\ a_1 a_2 = 8$

$\Rightarrow |a_1 - a_2| = 4$

$\Rightarrow |\Delta_1 - \Delta_2| = \frac{1}{2} \times 4 \sin 30^\circ \times 4 = 4$

51. 01225.00

Sol. $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$\Rightarrow \quad \frac{1}{2} = \cos\frac{\pi}{3} = \frac{b^2 + c^2 - a^2}{2bc}$

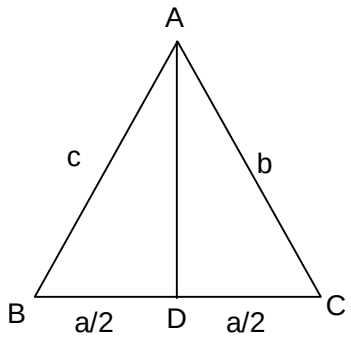
$\Rightarrow \quad b^2 + c^2 - a^2 = bc \qquad (1)$

In triangle ABD.

$AD^2 = c^2 + \frac{a^2}{4} - 2c \times \frac{a}{2}\cos B$

$\Rightarrow \quad 4AD^2 = 4c^2 + a^2 - 4ca\frac{(c^2 + a^2 - b^2)}{2ca}$

$= 2c^2 + 2b^2 - a^2$



$= 2c^2 + 2b^2 - \left(b^2 + c^2 - bc\right)$ [Using (1)]

$= b^2 + c^2 + bc$

$= 2500 + 900 + 1500 = 4900 \Rightarrow AD^2 = 1225$

52. 00002.00

Sol. Let the coordinate of A, B, C be $\left(t_i^2, 2t_i\right)$

$i = 1,2,3$ respectively. The tangent at A and B are $t_1y = x + t_1^2$ and $t_2y = x + t_2^2$ which intersect at $x = t_1t_2, y = t_1 + t_2$

So the vertices are $P\left(t_1t_2, t_1 + t_2\right)$, $Q\left(t_2t_3, t_2 + t_3\right)$ and $R\left(t_1t_3, t_1 + t_3\right)$

$\Delta = \left|\left(t_1 - t_2\right)\left(t_2 - t_3\right)\left(t_3 - t_1\right)\right|$

$$\Delta' = \left|\frac{1}{2}\begin{vmatrix} t_1t_2 & t_1 + t_2 & 1 \\ t_2t_3 & t_2 + t_3 & 1 \\ t_3t_1 & t_3 + t_1 & 1 \end{vmatrix}\right|$$

$$= \left|\frac{1}{2}\begin{vmatrix} \left(t_1 - t_3\right)t_2 & t_1 - t_3 & 0 \\ \left(t_2 - t_1\right)t_3 & t_2 - t_1 & 0 \\ t_3t_1 & t_3 + t_1 & 1 \end{vmatrix}\right|$$

$$= \left|\frac{1}{2}\left(t_1 - t_3\right)\left(t_2 - t_1\right)\left(t_2 - t_3\right)\right| \Rightarrow \Delta = 2\Delta'$$

53. 07225.00

Sol. Any point on the line is $r = \left(2\lambda + 1\right)i + \left(\lambda + 2\right)j + \left(2\lambda + 3\right)k$ which lies on the plane if $2\left(2\lambda + 1\right) - 6\left(\lambda + 2\right) + 3\left(2\lambda + 3\right) + 5 = 0 \Rightarrow \lambda = -1$ and the position vector of the point of intersection is $-i + j + k = ai + bj + ck$ $a = -1, b = 1, c = 1$

$\therefore \left(50a + 60b + 75c\right)^2 = \left(85\right)^2 = 7225$

54. 00009.00

Sol. $\lim\limits_{x \to 0^-} f\left(x\right) = \lim\limits_{h \to 0} f\left(0 - h\right)$

$$= \lim_{h \to 0} \frac{\sin^{-1}\left(1 - \left\{0 - h\right\}\right).\cos^{-1}\left(1 - \left\{0 - h\right\}\right)}{\sqrt{2\left\{0 - h\right\}.\left(1 - \left\{0 - h\right\}\right)}}$$

$$= \lim_{h \to 0} \frac{\sin^{-1}\left(1 - \left(1 - h\right)\right).\cos^{-1}\left(1 - \left(1 - h\right)\right)}{\sqrt{2\left(1 - h\right).\left(1 - \left(1 - h\right)\right)}}$$

$$= \lim_{h \to 0} \frac{\sin^{-1}h.\cos^{-1}h}{\sqrt{2\left(1 - h\right)}.h} = \frac{\pi}{2\sqrt{2}}$$